

Portfolio Case Study – Fedor Gorbunov

Analytics UI with Human-in-the-Loop Classification

Executive Summary

I designed and built a private analytics UI for reviewing market events, report outputs, candidate classifications, validation state, and manual review decisions. The goal was to make complex analytical results explainable and reviewable instead of showing only raw charts or opaque signals.

Business Problem

Raw charts and reports are not enough for serious analysis. Operators need to understand what was detected, why it matters, which parts were validated, which parts are still candidates, and where manual review overrides automatic output.

What I Built

The UI combines chart-based review, report cards, filters, KPI summaries, pending attempts, modal drill-down, event inspection, and human-in-the-loop classification. It separates manual truth, detector candidates, classifier output, validation state, and explanation layers.

Architecture Highlights

- Chart-driven review workflow.
- Backend-owned formulas and report calculations.
- Mongo-backed report scanner and event data.
- Day/symbol scoped chart state.
- Manual truth vs detector/classifier candidates.
- Modal drill-down for inspecting one event with metrics and visual context.
- Parameterized report builder for testing execution modes and rule variations.

Human-in-the-Loop Model

The UI separates what was manually reviewed, what was detected automatically, and what still needs validation. Manual review acts as the truth layer; automatic classification is treated as a candidate or audit layer until accepted.

My Role

Product Architect, Full-Stack Builder, Analytics UI Designer. I defined the workflow, designed the UI model, built the review surfaces, connected backend report logic, and shaped the validation/classification process.

Result

A review workflow where the operator can inspect events, compare candidates, track validation state, review KPIs, and understand why a case matters. The value is explainable analytics and decision support, not trading-profit claims.

Not Disclosed

Trading rules, thresholds, strategy parameters, proprietary calculations, account data, and private strategy logic.